

PID Control System Optimization Report

E-Town Rocket Bureau

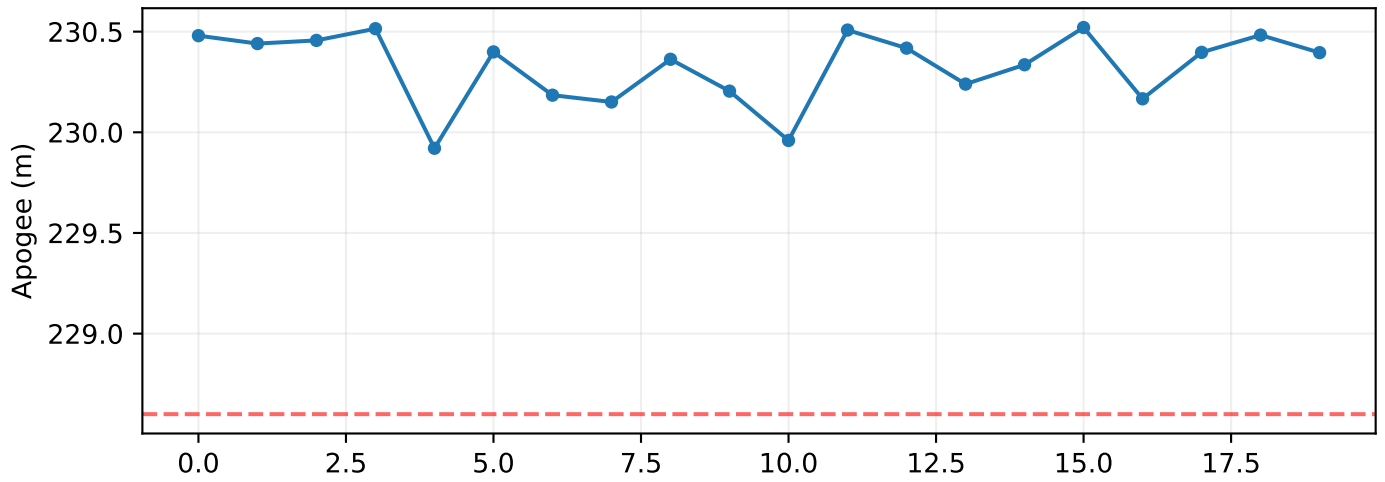
Generated on 2026-05-31 01:27:19

FINAL OPTIMIZED CONSTANTS

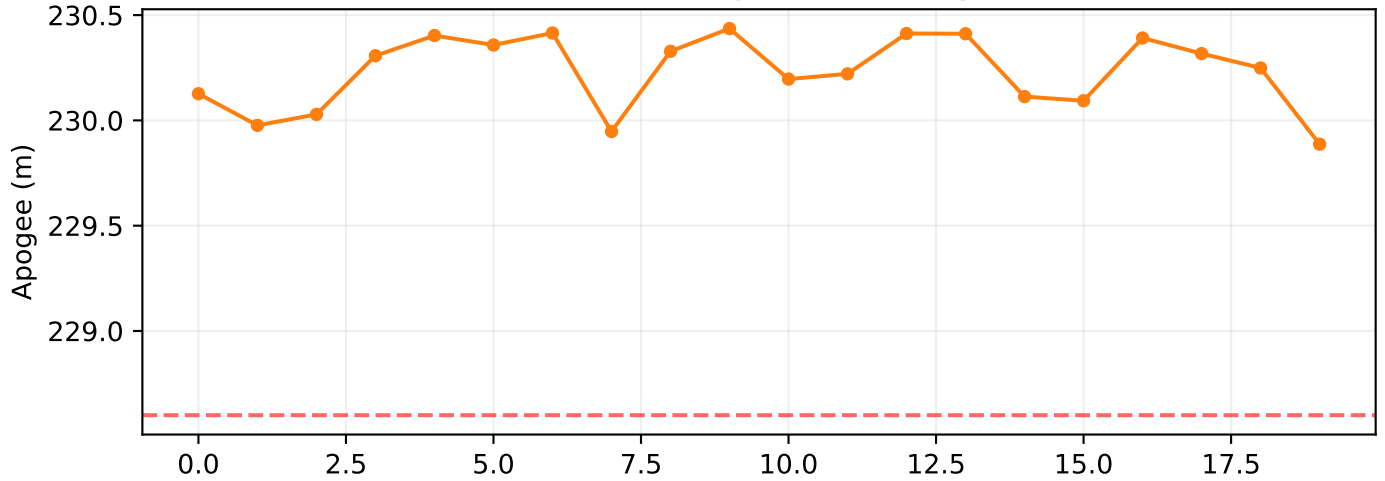
Kp: 0.0009999990
Kd: 0.0000100000
Ki: 0.0000999999

FINAL APOGEE: 230.0446 m
TOTAL SIM TIME: 51.6955s

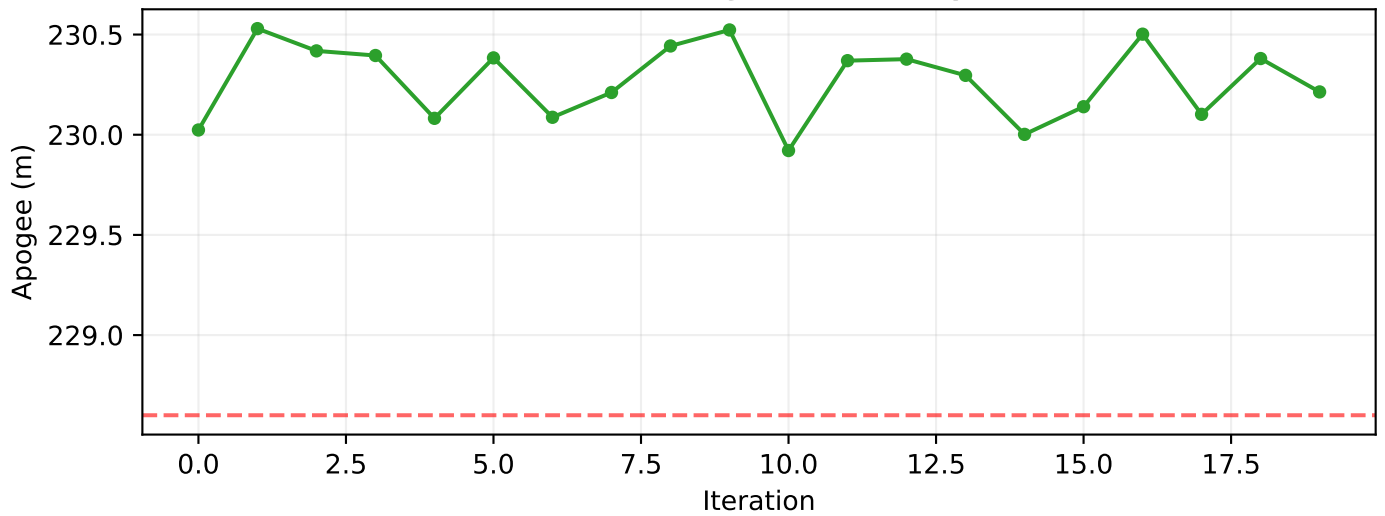
KP Convergence History



KD Convergence History



KI Convergence History



Kp Optimization Detailed History

Search Duration: 21.9930s

Iter	Value	Apogee (m)
1.0	0.0005005	230.4799819922
2.0	0.00075025	230.440698091
3.0	0.000875125	230.456470329
4.0	0.0009375625	230.5147132691
5.0	0.0009687812	229.9207095089
6.0	0.0009843906	230.3993596757
7.0	0.0009921953	230.1843925052
8.0	0.0009960977	230.1506051917
9.0	0.0009980488	230.3626920893
10.0	0.0009990244	230.2045908188
11.0	0.0009995122	229.9595921538
12.0	0.0009997561	230.5078356333
13.0	0.0009998781	230.4180189453
14.0	0.000999939	230.2393873946
15.0	0.0009999695	230.3355735913
16.0	0.0009999848	230.5203810362
17.0	0.0009999924	230.1664387256
18.0	0.0009999962	230.3971601448
19.0	0.0009999981	230.4829513258
20.0	0.000999999	230.3956740601

Kd Optimization Detailed History

Search Duration: 15.4404s

Iter	Value	Apogee (m)
1.0	5.005e-06	230.126741496
2.0	7.5025e-06	229.9764302917
3.0	8.7512e-06	230.0284214235
4.0	9.3756e-06	230.3071810406
5.0	9.6878e-06	230.4030371355
6.0	9.8439e-06	230.3581652208
7.0	9.922e-06	230.4149732947
8.0	9.961e-06	229.9474604974
9.0	9.9805e-06	230.3280966437
10.0	9.9902e-06	230.4360499392
11.0	9.9951e-06	230.1962402983
12.0	9.9976e-06	230.2210479873
13.0	9.9988e-06	230.4125702529
14.0	9.9994e-06	230.4115459337
15.0	9.9997e-06	230.1130156648
16.0	9.9998e-06	230.0933209
17.0	9.9999e-06	230.3913407631
18.0	1e-05	230.3172841239
19.0	1e-05	230.249018405
20.0	1e-05	229.8870295607

Ki Optimization Detailed History
Search Duration: 14.2493s

Iter	Value	Apogee (m)
1.0	5.005e-05	230.0236895888
2.0	7.5025e-05	230.5295861982
3.0	8.75125e-05	230.4184639637
4.0	9.37562e-05	230.3954117678
5.0	9.68781e-05	230.0818364909
6.0	9.84391e-05	230.3835082498
7.0	9.92195e-05	230.0870629589
8.0	9.96098e-05	230.2108179186
9.0	9.98049e-05	230.4429865089
10.0	9.99024e-05	230.5229855166
11.0	9.99512e-05	229.9211349915
12.0	9.99756e-05	230.3696984938
13.0	9.99878e-05	230.3772604991
14.0	9.99939e-05	230.2963341436
15.0	9.9997e-05	230.001807671
16.0	9.99985e-05	230.1398429238
17.0	9.99992e-05	230.5018873293
18.0	9.99996e-05	230.1017664767
19.0	9.99998e-05	230.3802340787
20.0	9.99999e-05	230.2136602379